

ECON 4970

Intro to Data Science for Economists

Summer 2026

This course will equip you with the ability to work with economic data and learn how to ask and answer questions using data and data analysis software. You will also learn to articulate technical findings to general audiences. This course will show you how to use, manipulate, and analyze large data sets in economics using data analysis software packages. This introductory course will provide you with foundations in data extraction, cleaning, and analysis to better prepare you for careers in economic data analyses and for advanced quantitative pursuits such as econometrics and research internships.

Course Goal & Objectives

By the end of this course you will be able to use data analyses tools to examine data and communicate key data insights to general audiences through mastering the following objectives:

1. Identifying the structure and sources of economic data
2. Reading and writing data using a variety of formats
3. Extracting, cleaning, and manipulating data for economic analysis
4. Detecting and managing missing data
5. Describing data, conducting basic analysis, and presenting results

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Class day/time: Tues & Thurs 1:50 – 4:20pm

Class location: 55 Park Place, Rm 601

Office Hrs: By appointment (email me)

Email me with your GSU email only as I am unlikely to receive emails that are not from official GSU student domains.

Performance & Evaluation

Here is an overview of assignments and their percentages of your grade:



Participation

(Class Attendance is Required)

10%



Labs/Assignments

(approximately 6 assignments which may include: dataset files, excel workbooks, Stata .do files, etc.)

40%



Data Project

Presentation

(5% for the project pitch, 15% final presentation)

20%



Data Project

(15% for code and 15% for report)

30%

Plus/Minus Grading Policy

This program uses a plus/minus grading scale. The scale and grades are:

98 - 100	A+	77 - 79	C+
93 - 97	A	73 - 76	C
90 - 92	A-	60 - 69	D
87 - 89	B+	0 - 59	F
83 - 86	B		
80 - 82	B-		

Required Materials

There is no required textbook. Lecture material will partly be based on material from Cameron, A.C. and P.K. Trivedi (2009), *Microeconometrics Using Stata* (Revised Edition), Stata Press. Other supplemental references, including instructional videos, demos, and other references will be shared via iCollege.

Personal Computer:

A personal laptop or desktop is highly recommended for this course. This course is designed to provide hands-on experience working with data and data analyses software. Although our classroom comes equipped with computers and software, it is recommended students have their own personal computers to work on assignments and projects outside of our regular class times. Students are encouraged to bring their own computers to class. If you do not have a personal computer, you can use computers at the GSU library or the AYSPS Econ Lab (additional details below). You also have the option to borrow equipment from CETL at:

(<https://library.gsu.edu/services-and-spaces/spaces-and-technology/borrow-equipment/>).

Computer Software:

1. **Stata (version 15.1 or higher)**

This course will introduce you to manipulating and analyzing data using Stata software. The computers in our classroom have Stata installed and ready for student use. Access to Stata software for personal computers is not included as part of students' technology fees, however, there are three ways you can go about using Stata this semester.

- 1) I highly recommend students purchase and download the student reduced-priced 6-month license for the latest version of Stata/BE (basic edition) on their own computer for \$94 available through the official Stata website:
<https://www.stata.com/order/new/edu/gradplans/student-pricing/>
- 2) Access Stata remotely using GSU's Virtual Lab. Free access is available through internet web browsing (HTML) or by downloading the Horizon Client (application-based).
<https://technology.gsu.edu/technology-services/it-services/labs-and-classrooms/vlab/>
Warning: There is a limit on how many students can use vLab at the same time.
- 3) Access Stata using on-campus computers at the GSU library or AYS Econ Labs (AYS 720 and 55 Park Place, room 601). Access to the labs is not available if there is another class currently in session at the lab.

2. **Microsoft Excel**

This course will provide a brief introduction to excel and cover some of the basic tools and features of Excel that can be used to describe and display data. Students can download and install the latest version of Microsoft Excel on their personal computer through the GSU Technology Services website free of charge by visiting the following site:

<https://technology.gsu.edu/technology-services/it-services/software-computer-purchase/software-download-and-purchase/>

Class Format and Attendance

This class meets face-to-face twice per week (Tuesdays & Thursdays from 1:50 to 4:20 pm). Attendance is required and will be collected during every class. In-class participation will count as 10% of your final course grade. Students are allowed 3 excused absences; after 3 the student will lose 1% of their course grade for every class missed.

Course Schedule & Assignments (subject to change)

Module & Section	What to Do	Due Date
Course Introduction		
Welcome & Course Introduction	- Read the course intro/overview information - Review Required Materials, Assignments, and Policies - Download and Review Syllabus	Jun 9
Module 1 – All About Data		
Data Basics & Finding Data	- Review materials in iCollege - Download data project instructions and make note of the various due dates - Start thinking of data project ideas - Start on Data Assignment 1	Jun 11
Module 2 – Introduction to Stata		
Stata Basics	- Review materials in iCollege - Submit Data Assignment 1 (due at noon) - Start on Data Assignment 2	Jun 16
Cleaning and Manipulating Data 1	- Review materials in iCollege - Lab Section 2 - Submit Data Assignment 2 (due at noon) - Start on Data Assignment 3	Jun 18
Cleaning and Manipulating Data 2	- Review materials in iCollege - Lab Section 2	Jun 23
Reorganizing and Aggregating Data	- Review materials in iCollege - Lab Section 3	Jun 25
Data Project Pitch	- In-class data project pitch presentations (5-min) - Start on Data Assignment 4	Jun 30
Analyzing and Presenting Data Part 1	- Review materials in iCollege - Lab Section 4 - Submit Data Assignment 3 (due at midnight)	Jul 2
Analyzing and Presenting Data Part 2	- Review materials in iCollege - Lab Section 5	Jul 7
Analyzing and Presenting Data Part 3	- Review materials in iCollege - Lab Section 6 - Submit Data Assignment 4 (due at midnight)	Jul 9

Module & Section	What to Do	Due Date
	Start on Data Assignment 5	
Module 3 – Other Data Analysis and Visualization Tools		
Excel/R Studio Workshop Part 1	Work on Data Project	Jul 14
Excel/R Studio Workshop Part 2	- Work on Data Project - Submit Data Assignment 5 (due at midnight) - Start on Data Assignment 6	Jul 16
Impact Evaluation	Practical Example of How to Evaluate a Public Policy	Jul 21
Data Project Presentations and Project		
Presentation Day	10-minute presentation with data visuals - Submit Data Assignment 6 (due at midnight)	Jul 23
Final Data Project	Written data appendix report + programming code by 11.59pm	Jul 28

Late Assignment Policy

Coding assignment solutions will be posted shortly after the due date so **late assignment submissions will not be accepted**. Late submissions on Final Projects and Presentations will be penalized 10% for each day late.



Helpful Notes from Your Professor

Here are some important notes on how you can be successful in this class:

✓ Embrace Uncertainty

This class involves project work that can be messy, nonlinear, and evolving. Your approach to a coding assignment may differ from another classmate's or the professor's strategy. You won't get everything right the first time, but keep trying and adapt as you learn..

✓ Learn to be Resourceful

Stata is a programming tool that is used by millions of economists and other data analysts. The problems that you will encounter likely have been asked and answered already by other users through online platforms like Stata.com, YouTube, and research blogs. Learn to use the web to troubleshoot your issues including our iCollege discussion board.

✓ Manage Your Time

This course is face-to-face and I plan to use some classtime for helping with assignments and projects. Coding is not always straightforward and sometimes it can take a while to get help on issues you are stuck on. Make sure you set aside at least 6 hours per week to complete your work. You will have assignments due most by 12-noon before class.

✓ I'm Here to Help

Don't hesitate to contact me with questions or problems at any point during the semester either in class or via email. I may not have all the answers, but I can help you figure out how best to ask the right questions to find a solution to your problem

Support Statements

Inclusivity Statement

We understand that students in our program come from a variety of backgrounds and perspectives. AYSPS is committed to providing a learning environment that respects diversity. To build this community we ask all members to:

- Share their unique experiences, values and beliefs
- Be open to the views of others
- Honor the uniqueness of their colleagues
- Appreciate the opportunity that we have to learn from each other in this community
- Value each other's opinions and communicate in a respectful manner
- Keep confidential discussions that the community has of a personal (or professional) nature

Students with Disabilities

Students who wish to request accommodation for a disability may do so by registering with the GSU Access & Accommodations Center (AACE). Students may only be accommodated upon issuance of a signed Accommodation Plan by AACE. Students are responsible for providing a copy of that plan to instructors of all classes in which an accommodation is sought.

To register for accommodations please follow this link <https://access.gsu.edu/how-to-connect/how-to-register/>

For more information, contact AACE located at Student Center East, Suite 205, 55 Gilmer Street, Atlanta, GA 30303. Phone: 404-413-1560 Email: access@gsu.edu

Remote Academic Coaching

The Office of Disability Services also offers free remote academic coaching. To learn more about these services go to disability.gsu.edu/services or watch a [Coaching Video](#).

Veterans & Serving Military

Georgia State honors its military and veteran men and women returning to pursue their education. Students who are veterans, serving in the military, and their dependents are encouraged to avail themselves of a full range of college services and activities through the Military Outreach Center (MOC).

For assistance or guidance while attending GSU on campus or online, contact the Atlanta Campus Military Student Advocate, Randy Barrone at 404-413-2331. Also, please be sure and let me know ASAP if or when there is any possibility of you being activated and deployed. Thank You For Your Service!

For more information contact the GSU Military Outreach Center
Phone: (404) 413-2331, Email: rbarrone@gsu.edu, Website: military.gsu.edu
Address: Dahlberg Hall, 30 Courtland Street, Suite 217, Atlanta, GA 30303

Online Course Evaluations – Student Surveys

Your constructive assessment of this course plays an indispensable role in shaping improvements of all courses within this program and your educational experiences at Georgia State. Please take time to fill out the online course evaluations. We appreciate honest and open feedback.